CAT: Center Attention Transformer With Stratified Spatial–Spectral Token for Hyperspectral Image Classification

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***Abstract*— Most hyperspectral image (HSI) classification methods rely on square patch sampling to incorporate spatial information, thereby facilitating the label prediction of the center pixel. However, square patch sampling introduces numerous heterogeneous pixels, which could distort the label prediction of the center pixel. Moreover, it generates fixed training patch sample for each center pixel, hampering the performance of transformer-based models requiring a large number of training data. To address the above problems, we proposed center attention transformer (CAT) with stratified spatial–spectral token generated by superpixel sampling for HSI classification. First, to mitigate the inference of heterogeneous pixels, we propose sampling from superpixel region (SSR) mechanism to generate purer image cubes than traditional square neighborhood. Second, to expand the training data for transformer, we propose multiple stratified random sampling (MSRS) mechanism, which generates ample training samples without introducing additional labels. Finally, to more effectively extract information from the sampled patch tokens, we propose spatial–spectral token generation mechanism and CAT structure with Gaussian positional embedding (GPE). This framework can extract long-range correlations of spectral information and pay more attention on the center pixel in spatial dimension. Experimental results on three HSI datasets demonstrate the performance of our proposed method CAT outperforms several state-of-the-art methods. The code of this work is available at https://github.com/fengjiaqi927/CAT-Center_Attention_Transformer.**

***Index Terms*— Hyperspectral image (HSI) classification, multiple random sampling, positional embedding super pixel segmentation, transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) is composed of hundreds of continuous spectral bands, which contain rich spectral information. Each band characterizes the spatial distribution of the ground radiation or reflection at the corresponding wavelength. Each pixel corresponds to a 1-D spectral characteristic curve, which contains the unique properties of the pixel region. This characteristic makes it possible for pixel-level classification. HSI classification task is the foundation of hyperspectral remote sensing applications, such as mineral exploration [1], precision agriculture [2], [3] and military reconnaissance [4].

In the last few years, convolutional neural networks (CNNs), as a powerful approach while dealing with image processing tasks, have achieved great success in HSI classification. Different from early stage traditional classification approaches [5], [6], [7], such as support vector machines (SVMs) [5], Random Forest (RF) [6] and K-nearest-neighbor (KNN) [7], CNN enhances the translation invariance and local inductive bias of the network through locally shared convolution kernels [8], [9], [10], [11]. This mechanism fully exploits the local spatial relationship of HSI through square patch-wise input image cubes, realizing deep-level spatial–spectral information fusion. By taking the lead, Chen et al. [8] proposed a CNN-based feature extraction model to extract nonlinear, discriminant, and invariant spectral–spatial features of HSI. With the powerful feature extraction capability of the network, this method achieves more compelling results than traditional methods. Since then, the fixed square region cropped from HSI has become a conventional input for CNN-based methods. This cube data contain spectral information of both the central pixel to be classified and the spatial context area. Based on this consensus, scholars have designed various structures for mining HSI cube features efficiently and discriminatively. To relieve the curse of dimensionality, Santara et al. [9] proposed a deep network architecture called BASS. The network consists of a band selection and fusion block, a spectrum joint feature learning block, and a classification block, which is able to extract band-specific spectral–spatial features. To extract spatial and spectral features independently, Hao et al. [10] proposed a two-stream architecture. One stream employs the stacked auto-encoder to extract spectral information and the other stream employs the CNN to process the corresponding image patch.

Benefiting from the spatial smoothness assumption, these CNN-based networks can well aggregate features from local homogeneous regions. However, the assumption of spatial smoothness does not apply at the class boundary. As shown in Fig. 1, when the pixel to be classified is at the class

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boundary, the patch-wise input contains a large number of heterogeneous pixels. The overlap of input features among adjacent heterogeneous samples can indeed lead to confusion during classification.

To solve this problem, some scholars have proposed methods based on weight redistribution [12], [13], [14], [15], [16], [17], [18]. The core idea of these methods is to re-weight the features through the attention mechanism, which weakens the weight of heterogeneous regions and strengthens the weight of distinguishable features. To suppress the effects of heterogeneous pixels, Sun et al. [12] proposed a spectral–spatial attention network (SSAN) to capture discriminative features from HSI cubes. Zhao et al. [13] adopt a center attention module (CAM) that pays more attention to the features that are more correlated with the target pixel. Huang et al. [14] proposed a two-branch attention adversarial domain adaptation (TAADA) network and adjust the distribution between the source and target domains to extract domain-invariant features. In addition, some scholars have proposed spectral attention for HSI classification. For instance, Roy et al. [15] proposed a fused squeeze-and-excitation network (SENet), which fuses the excitation scores from multiple squeeze channels and prioritizes the important channels toward better convergence. Zheng et al. [16] design a center spectral attention (CSpeA) module to avoid the influence of other categories of pixels to suppress redundant spectral bands.

Different from the attention mechanism, some scholars have considered combining traditional superpixel segmentation algorithms [19], [20], [21] with CNN to guide the network to focus on homogeneous regions. Simple linear iterative clustering (SLIC) [19] is one of the most common algorithm that is used in HSI processing. This algorithm performs a local regional clustering iteratively by incorporating the spatial correlation and the spectral similarity simultaneously. As a result, this method can cluster similar pixels into the same superpixel region. Liu et al. [22] considered that the superpixel-based constraint can be dedicated to the spatial homogeneity and pseudolables adaptivity. Li et al. [23] proposed a homogeneous region generated by SLIC to instruct the network to learn information in irregular region. Zhao et al. [24] proposed a novel superpixel guided deformable convolution network (SGDCN) to extract pure neighborhood spatial–spectral features. By guiding the CNN-based structures with the superpixel maps, these methods can indeed concentrate on the target regions and ignore the effects of interfering pixels intuitively.

Furthermore, some scholars have introduced graph convolutional networks (GCNs) into CNNs. Compared with the structural limitations of CNNs, GCNs can flexibly select graph nodes, making it possible to eliminate the interference of heterogeneous samples. Gong et al. [25] proposed a superpixel spectral–spatial feature fusion graph convolution network to learn the spatial information under different spatial neighborhoods. Mou et al. [26] proposed an end-to-end nonlocal graph convolutional network, which reduces the heterogeneity of the input data by Laplacian smoothing. Nonetheless, the graph nodes are selected from the whole map and it is computationally complex. Hong et al. [27] introduced GCN

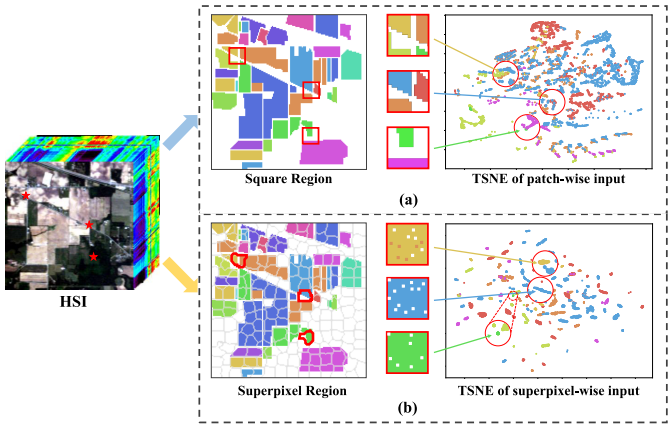

Fig. 1. Feature visualization of (a) patch-wise input and (b) superpixel-wise input using TSNE [28]. Each point represents an input image cube, and the color represents the corresponding category.

to capture long-range spatial relationships between samples. Although the sample selection is flexible, it introduces inherent defects of CNN cause the network structure is composed of GCN and CNN.

In general, the introduction of attention mechanism, superpixel segmentation algorithms, and GCN can effectively redistribute the weight of input samples and reduce cubes heterogeneity, improving the classification accuracy. However, such methods still have their inherent flaws. First, the receptive field of CNNs is local. It is difficult for shallow layers to pay attention to the long-distance dependencies of samples. Second, the spectral features are aggregated by weighted summation, which does not mine the correlation between bands. At last, the above methods only redistribute the weights, which just weaken the interference of heterogeneous samples instead of eliminating them fundamentally.

Very recently, the attention-based transformer framework has drawn intensive attention from the research community. This structure is capable of processing sequence data and mining long-range relationships [29], [30], [31], [32], [33], [34], [35]. In order to explore the potential of transformers in hyperspectral classification tasks, He et al. [29] directly flattened the HSI cube into pixel sequences and fed all pixels into HSI-BERT to extract relations among them. Benefiting from the flexible sampling capability of this structure, He designed a variety of token collection methods. However, the manual setting regions are difficult to generalize the complex distribution of ground features and the design of the network structure also lacks adaptability. Since then, many scholars have made innovations in structure design to enhance the adaptability to HSI processing. To extract the long-range spectral correlation, Hong et al. [36] proposed SpectralFormer to split the spectra into tokens. To capture the fine-grained spatial information as well as establish interdependencies between spectral sequences, Peng et al. [37] proposed a spatial–spectral transformer with cross-attention (CASST) method. Wang et al. [38] proposed a novel bilateral classification network to fully exploit the abundant spatial information and spectral correlations. Sun et al. [39] proposed a spectral–spatial feature tokenization transformer (SSFTT)

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method to capture high-level semantic features. Mei et al. [40] proposed the group-aware hierarchical transformer for HSI classification. They employed pixel embedding instead of patch embedding to encode hyperspectral cubes and feed them into the transformer structure. Qiu et al. [41] proposed a cross-channel dynamic spatial–spectral fusion Transformer (CDSFT) to describe the cross-channel dependencies. Nevertheless, most of these improvements are from the perspective of feature extraction. There is no extend research on data sampling design following the previous work of He et al. [29]. Therefore, the flexible sequence selection advantage of transformer has not been fully utilized. The conventional square sampling can lead to a mix of heterogeneous pixels at category boundaries, which disrupt the classification of the central pixel for transformers. Moreover, the conventional square sampling provides only a limited number of fixed patch samples, which restricts the training potential of the transformers. Unlike the inductive biases present in convolutional networks, such as translation invariance and local receptive fields, the transformer architecture makes few assumptions about structural information of data. This makes transformer a universal and flexible architecture. As a side effect, the lack of structural bias makes transformer prone to overfitting for small-scale data [42]. According to the experiments in [43], the transformer has been shown to require a large amount of data for training which contradicts the limited hyperspectral data.

In order to eliminate the heterogeneity in the input samples fundamentally, we take advantage of the spatial sampling flexibility and propose sampling from superpixel region (SSR) strategy. To alleviate the over-fitting phenomenon in transformers, we expand the training data without additional labels and reducing the model complexity by designing a corresponding inductive bias for HSI classification. We propose a CAT with stratified spatial–spectral token for HSI classification. This framework achieves coarse samples division through superpixel segmentation, as shown in Fig. 1, the input data purity has been improved with the help of stratified random sampling in superpixel region. At the same time, multiple sampling rule can generate a large number of data for the training process of transformer. In addition, the designed Gaussian positional embedding (GPE) can enhance the attention to the center pixel to be classified. The main contributions of this article can be summarized as follows.

1. We propose SSR module for transformer, which eliminates the influence of heterogeneity and improves the data purity. This method reduces the introduction of heterogeneity at category boundaries.

2. We design MSRS module for training data generation, which generates a large number of training samples without adding labels. This rule alleviates the overfitting problem in transformer.

3. We propose a CAT framework by amalgamating spatial–spectral token with GPE token. The module can extract long-range correlations of spectral information and pay more attention on the center pixel in spatial dimension.

The remaining of this article is organized as follows. Section II gives a brief illustration of our method. Section III illustrates three benchmark HSI datasets, experiments settings, results, and discussions. Section IV draws conclusions of the current work and takes a brief outlook of the future research work.

II. Our Method

In this section, we will illustrate the essential components of the proposed method. The overall architecture of the method proposed in this article is shown in Fig. 2. We first introduce the application of superpixel segmentation in HSI classification. On this basis, we propose a date generation rule by SSR instead of original square region to improve data purity. Then we propose MSRS strategy without extra labels to alleviate the overfitting problem at the data level. In addition, we proposed a Spatial–Spectral Token Generation strategy to extract long-range discriminative information both in spatial and spectral dimensions. At last, we illustrate the proposed CAT structure with GPE, which introduces structural prior information and effectively alleviates overfitting problem at the structural level.

A. Sampling From Superpixel Region

As mentioned above, the current deep learning-based frameworks are square patch-based frameworks. The patch consists of two elements: the center pixel with a label or the one to be classified, and its neighboring pixels. Suppose the HSI dataset as $X \in R^{n\times d}$. It can be represented as follows:

$$X = \{x_1, x_2, \ldots, x_i, \ldots, x_n\}, \quad n = H \times W, \ x_i \in R^d. \quad (1)$$

Suppose the conventional patch generated from the square region as

$$P_i' = \left\{ x \in X \mid MD(x, x_c) \le \sqrt{M} \right\} \quad (2)$$

where P_i' is the ith conventional patch and x_c is the center pixel of P_i'. MD() is the Manhattan Distance of two pixel positions. M is the patch size.

Different from generating the patch from the square region, we generate the patch by SSR to improve the data purity. The superpixel segmentation algorithm we used is SLIC [19]. The SLIC algorithm can divide the HSI dataset into multiple superpixel regions. The kth superpixel segmentation region $C_k \in R^{n_k\times d}$ is formulated as follows:

$$C_k = \left\{ x \in X \mid \mathrm{dist}\left(x - x_{C_k}\right) < T \right\}, \quad k = 1, 2, \ldots, K \quad (3)$$

where x_{C_k} is the cluster center of C_k and T is the clustering threshold. The $dist()$ is the metric function in SLIC [19].

Then we generate the supposed patch by SSR and the ith patch P_i is formulated as follows:

$$P_i = \{x \in C_k \mid \mathrm{card}(P_i) = M\} \quad (4)$$

where $\mathrm{card}(P_i)$ is the number of elements in the set P_i and M is the patch size. In the patch, the pixel with a label or the one to be classified belong to the superpixel region C_k. Pixels in our superpixel region patch P_i have similar spectrum, while pixels in conventional square region patch P_i' are only spatially adjacent. This means that when spatially adjacent pixels do

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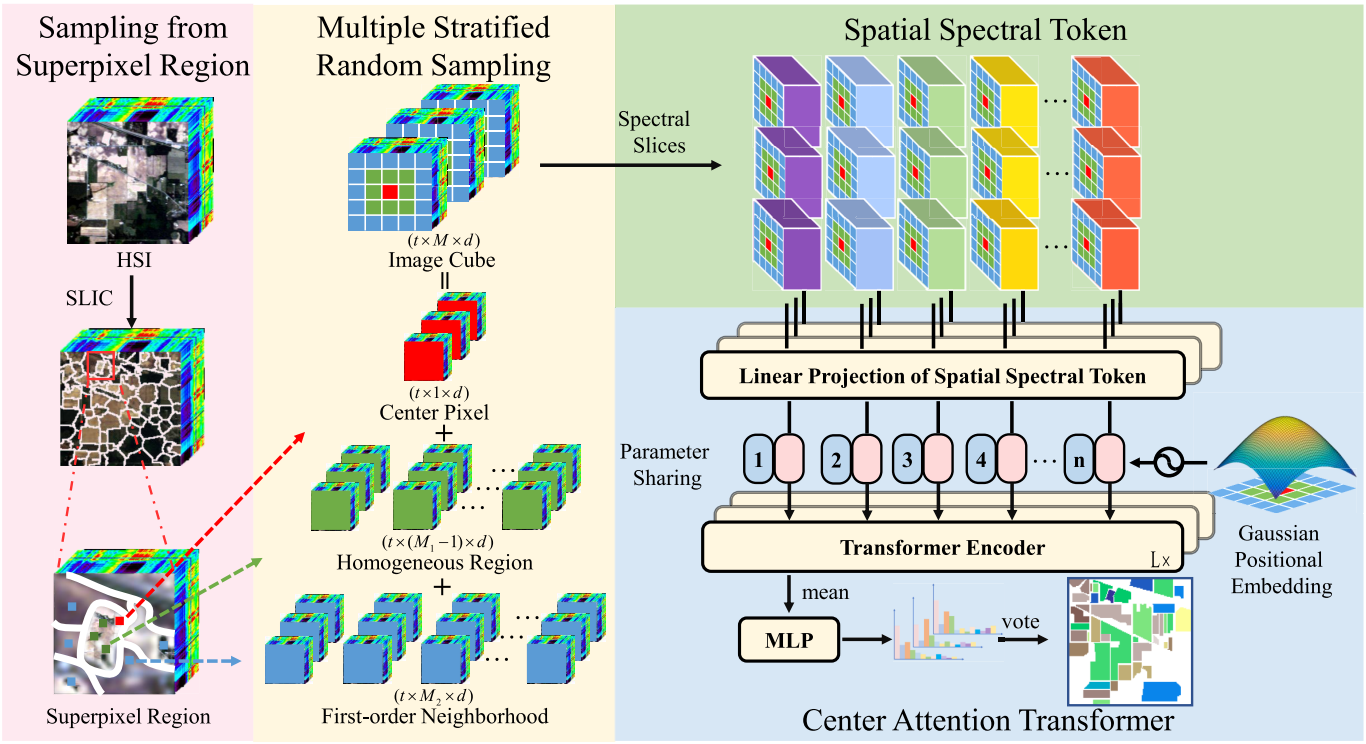

Fig. 2. Overall framework of the proposed CAT, which consists of four blocks. First, the SSR block utilizes SLIC algorithm to segment the input HSI into superpixel map. Second, the MSRS block generates image cube by sampling from specific regions randomly, which consists of center, homogeneous and neighborhood region. Third, the spatial–spectral token generation block split the image cube into spatial–spectral token. Ultimately, the CAT block utilizes transformer encoder with GPE to extract discriminative features.

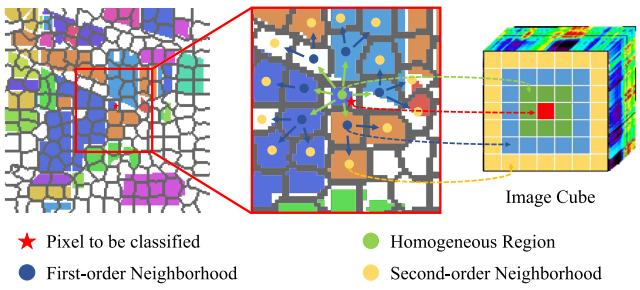

Fig. 3. MSRS module. To classify the pixel, we generate multiple image cubes by sampling from homogeneous region and neighborhood region randomly instead of cropping square patch. The image cubes are independent and identically distributed. The ablation study of different sampling regions is shown in Table III.

not belong to the same class, the consistency of elements in superpixel region patch will be stronger than square region patch. This phenomenon often occurs at the boundary of categories, so our generation method can effectively reduce the interference of heterogeneous noise at the boundary.

B. Multiple Stratified Random Sampling

To alleviate the overfitting problem in transformer, we proposed MSRS rule to generate a large number of training samples without adding labels.

When setting different number of superpixels K in SLIC algorithm, the produced segmentation results have different spatial scales, which could affect the sampling region. A larger K-value results in a more refined clustering result, where the first-order neighborhood is devoid of heterogeneous pixels. Conversely, a smaller K-value yields a coarser clustering result, potentially including heterogeneous pixels within the first-order neighborhood.

In order to weaken the influence of the number of superpixels K, we expand the sampling area into a multiorder neighborhood superpixel region.

The specific division rules are as shown in Fig. 3. We first select the center superpixel where the labeled or to be classified pixel belongs to. Then we seek for the first-order neighborhood regions around the central superpixel and so on for higher-order regions. Points sampled from homogeneous regions are strategically positioned in the central area of the cube, while those from neighboring regions are situated at the cube's periphery. This spatial arrangement ensures that even if the first-order neighborhood encompasses heterogeneous pixels, their impact is attenuated by the introduced GPE in Section II-D.

Taking the first-order case as an example, we generate patch P_i from central superpixel C_k and first-order neighborhood superpixels $C_{ki}, i = 1, 2, \ldots, n$. Then the patch we generate is formulated as follows:

$$\begin{aligned} P_i &= A \cup B \\ A &= \{x \in C_k \mid \text{card}(A) = M_1\} \\ B &= \{x \in C_{k1} \cup C_{k2} \cup \cdots \cup C_{kn} \mid \text{card}(B) = M_2\} \end{aligned} \tag{5}$$

where M_1 and M_2 are the numbers of selected pixels and $M = M_1 + M_2$. Taking a 7×7 patch as an illustrative example, 7×7-5×5 pixels from the first-order neighbors

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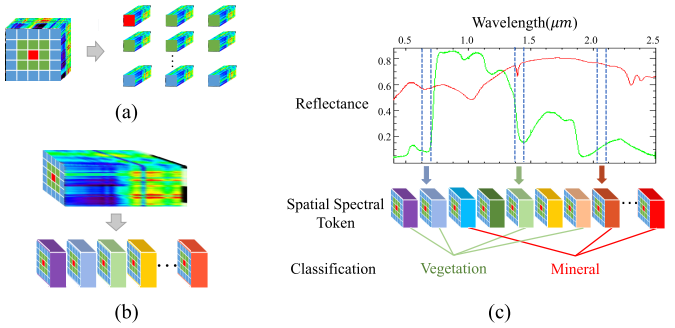

Fig. 4. Comparison of token generation methods and advantage analysis of spatial–spectral token in transformer-based framework. (a) Spatial token generation. (b) Spatial–spectral token generation. (c) Token correlation mining.

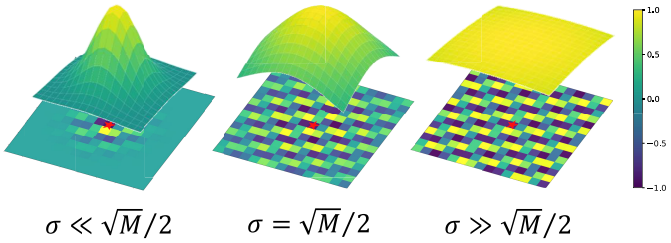

Fig. 5. Weights distribution of the GPE with different hyperparameter variance σ.

and 5×5-1 from homogeneous region. Additional dimensions and their corresponding sampling rules can be referred to in the released code. By setting different sampling ratios in a stratified manner, we can get a large number of training patches. Multiple random sampling ensures that the generated patches are independent and identical distribution. Since the data augmentation does not change the spectral characteristics, this method does not increase the feature space distribution of ground objects and avoids the influence of spectral distortion.

C. Spatial–Spectral Token Generation

To extract long-range discriminative information both in spatial and spectral dimensions, we propose a spatial–spectral token split mechanism. The specific split mechanism and its advantage are as shown in Fig. 4.

In HSI processing, there are two ways to divide a patch into tokens [37]. One is to divide the patch into spatial tokens in the spatial dimension, that means a single pixel is a token. Another is to divide the patch into spatial–spectral tokens in the spectral dimension. Different from the former one, Each spatial–spectral token is composed of multiple adjacent bands, which contains both spatial information and adjacent spectral information. It is worth noting that when the number of adjacent bands is equal to zero, the spatial–spectral token degenerates into spectral token. These tokens encourage the transformer-based methods to mine the spectral correlation of different grounds and improve the classification accuracy.

In this article, we use the second method to divide the patch into tokens. Suppose the patch generated by the way in Section B as $P_i = \{x_{i,1}, x_{i,2}, \ldots, x_{i,M}\} \in R^{M\times d}$. We generate token sequences $\{T_1, T_2, \ldots, T_k, \ldots\}$ by split $P_i^T = \{s_{i,1}, s_{i,2}, \ldots, s_{i,d}\} \in R^{d\times M}$. Then the token T_k can be formulated as follows:

$$T_k = \left\{ s_{i,c\times(k-1)+1}, s_{i,c\times(k-1)+2}, \ldots, s_{i,c\times k} \mid s \in P_i^T \right\} \tag{6}$$

where c is the number of adjacent bands and $s_{i,j} \in R^M$ is the jth band. Each patch will be divided into $\lceil (d/c) \rceil$ tokens. When d is not divisible by c, we will padding the patch by copying the last band until the new patch can be divided. The reasons for this setting are twofold. On the one hand, the spatial–spectral tokens can be used in transformer structure for autocorrelation calculation. This method can obtain the long-range dependence between spectra and effectively mine the information both in spatial and spectral dimensions. On the other hand, this method has no limitation on the spectral dimension. The change in the number of tokens will not affect the design of the network structure. Subsequently, this setting has potential scalability to process inputs of different spectral dimensions.

D. Center Attention Transformer

To extract long-range correlations of spectral information and pay more attention on the center pixel in spatial dimension, we propose a parameter-sharing network based on transformer structure called CAT. This network is composed of three parts: linear projection, the proposed GPE, and self-attention mechanism.

The spatial–spectral tokens are passed through a linear projection layer, which is formulated as follows:

$$T_i' = WT_i. \tag{7}$$

Subsequently, a novel positional embedding method called GPE is proposed to stress the importance of the central pixel to the classification of image cube as well as distinguish different tokens. In the GPE method, each pixel in the image cube is assigned a Gaussian distribution based on its distance from the center pixel. The closer a pixel is to the center pixel, the higher its probability density in the Gaussian distribution. The spatial–spectral tokens and GPE tokens are concatenated along the channel dimension to create the input image cube representation. By emphasizing the importance of the central region, the extracted feature contains critical information for classification tasks. The proposed GPE is formulated as follows:

$$\begin{aligned} \mathrm{GPE}_{(i,2\ j)} &= \sin\left(i/\alpha^{2j/c_0}\right) \times G_j \\ \mathrm{GPE}_{(i,2\ j+1)} &= \cos\left(i/\alpha^{2j/c_0}\right) \times G_j \end{aligned} \tag{8}$$

where i is the band number, j is the pixel number, and we let $\alpha = 10\ 000$ as in [43]. G_j is the jth Gaussian weight which is formulated as follows:

$$G_j = e^{-\left((x-x_0)^2+(y-y_0)^2\right)/(2\sigma^2)} \tag{9}$$

where (x_0, y_0) is the center pixel coordinate. (x, y) is the jth pixel coordinate which is calculated as follows: $x = \lfloor j/w \rfloor$, $y = j - x \times w$. w is the width of patch and σ is a hyperparameter which controls amplitude.

The GPE weight distribution is shown in Fig. 5. When hyperparameter σ is much larger than the patch size

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$((M)^{1/2})/(2)$, the position embedding falls back to the Sinusoidal Encoding [43]. Although Sinusoidal Encoding method realizes the relative position representation of the spectral dimension and long-range dependent attenuation, it lacks the adaptive adjustment to the spatial distribution of the HSI. When hyperparameter σ is much smaller than the patch size $((M)^{1/2})/(2)$, the positional encoding values tend to 0 at edges. The position embedding only focuses on the central region, which cannot be used to distinguish the position of boundary pixels. In order to comprehensively consider the influence of Gaussian amplitude and Sinusoidal Encoding on spatial and spectral dimension, we let $\sigma = ((M)^{1/2})/(2)$ and subsequent experiments will prove the superiority. In this case, GPE enables central region attention, relative position representation, and long-range dependence decay.

As mentioned in Fig. 4, the transformer encoder with self-attention mechanism allows the network to capture long-range relationships among the spectral tokens.

Multihead self-attention (MSA) is the core component of transformer encoder block and self-attention mechanism is as follows:

$$z = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{10}$$

where d_k is the dimensions of K. Each matrix in (10) is obtained by a linear mapping of the input sequence. In the first layer of the transformer encoder, we represent the input matrix as follows:

$$\begin{aligned} Q &= \left(T_i' + \text{GPE}\right)W^Q \\ K &= \left(T_i' + \text{GPE}\right)W^K \\ V &= \left(T_i' + \text{GPE}\right)W^V. \end{aligned} \tag{11}$$

We send patches to the transformer encoder block [43] with MSA to extract long-range feature of the spectral bands. Then the high-level features extracted by the transformer encoder will be typically pooled into a single vector and passed through a multilayer perceptron (MLP). The MLP learns to map the extracted features to class probabilities of the input image cube. Finally, an integrated learning method is used to produce the final classification result.

Overall, the CAT architecture is a powerful approach for HSI classification tasks that require the extraction of long-range correlations in spectral dimension and the emphasis on the central pixel in spatial dimension. The transformer encoder layers combined with GPE allow the network to capture complex relationships among the input tokens and extract discriminative features that can be used for classification or other downstream tasks.

III. Experiments and Analysis

A. Data Description

To verify the effectiveness of our proposed sampling strategy and CAT architecture, extensive experiments have been implemented on three widely used HSI classification datasets, i.e., Indian Pines (IP) [44], Pavia University (PU) [45], and Houston 2013 [46].

1) IP: The IP dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over North-Western Indiana, USA. It comprises 145×145 pixels with a 20-m spatial resolution and 224 spectral bands within the 0.4×10^{-6} to 2.5×10^{-6} meter wavelength range. After the removal of 24 water absorption bands, it retains 200 bands covering 16 classes of land features.

2) PU: PU dataset was collected by using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the city of Pavia, Italy. The image consists of 610×340 pixels with a 1.3 m spatial resolution and 103 spectral bands in the wavelength range 0.38×10^{-6} to 0.86×10^{-6} m. The image is divided into nine classes with total of 42776 labeled samples.

3) Houston 2013: Houston 2013 dataset was acquired by the NSF-funded Center for Airborne Laser Mapping (NCALM) over the campus of the University of Houston and its neighboring rural areas, USA. The image consists of 349×1905 pixels with a 2.5 m spatial resolution and 144 spectral bands in the wavelength range 0.38×10^{-6} to 1.05×10^{-6} m. The image is divided into 15 classes with total of 15 029 labeled samples.

To evaluate the performance on HSI classification tasks, we randomly partition three typical HSI datasets in accordance with Table I. The quantities of training and testing samples are aligned with those defined in previous studies [36]. This standard setting allows us to contrast the performance of different algorithms and derive credible conclusions about their effectiveness in HSI classification tasks.

B. Experimental Setup

In order to demonstrate the classification performance, we employ three common indices, *i.e.*, *Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient*. *OA* refers to the percentage of correctly classified samples in the classification task, while *AA* represents the average accuracy achieved across different classes. The Kappa coefficient denotes the percentage of classified pixels corrected by the number of agreements that would be expected purely by chance. Additionally, we provide visual results of the classification map to intuitively convey insights about the relationship between the algorithm accuracy and spatial distribution of classes in HSIs. This qualitative comparison gains a better understanding of the algorithm's performance and identify any patterns or areas where improvements may be needed.

The configurations of the proposed network CAT are detailed below. We choose 11th, 21st, and 43rd bands as convention to generate false color composite. Before SSR, we use SLIC [19] to segment all channel HSI. In order to balance the internal consistency of superpixel regions and the sampling range of receptive fields, we set the number of cluster centers $K \approx \alpha \times (H \times W)/(M_1)$. H and W are the image size. M_1 is the numbers of selected pixels in central superpixel and we set $M_1 = 15 \times 15$. α is a hyperparameter and we set $\alpha = 2$. In IP dataset, $K = 200$. In PU dataset, $K = 2000$. In Houston 2013 dataset, $K = 6000$. In MSRS module, we sample three times from the central superpixel region and its first-order neighborhood. In Spatial–Spectral Token Generation module, the number of adjacent bands $k = 3$. In CAT

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TABLE I
NUMBER OF RANDOMLY SELECTED PIXELS FOR THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASET

Class	Indian Pines			Pavia University			Houston 2013		
	Name	Train	Test	Name	Train	Test	Name	Train	Test
1	Alfalfa	15	39	Asphalt	548	6304	Healthy grass	198	1053
2	Corn-notill	50	1384	Meadows	540	18146	Stressed grass	190	1064
3	Corn-mintill	50	784	Gravel	392	1815	Synthetic grass	192	505
4	Corn	50	184	Trees	524	2912	Trees	188	1056
5	Grass-pasture	50	447	Painted metal sheets	265	1113	Soil	186	1056
6	Grass-trees	50	697	Bare Soil	532	4572	Water	182	143
7	Grass-pasture-mowed	15	11	Bitumen	375	981	Residential	196	1072
8	Hay-windrowed	50	439	Self-Blocking Bricks	514	3364	Commercial	191	1053
9	Oats	15	5	Shadows	231	795	Road	193	1059
10	Soybean-notill	50	918	-	-	-	Highway	191	1036
11	Soybean-mintill	50	2418	-	-	-	Railway	181	1054
12	Soybean-clean	50	564	-	-	-	Parking Lot 1	192	1041
13	Wheat	50	162	-	-	-	Parking Lot 2	184	285
14	Woods	50	1244	-	-	-	Tennis Court	181	247
15	Buildings-Grass-Trees-Drives	50	330	-	-	-	Running Track	187	473
16	Stone-Steel-Towers	50	45	-	-	-	-	-	-
	Total	695	9671	Total	3921	40002	Total	2832	12197

structure, We utilize a single-depth transformer block with 16 multihead attention mechanisms which share parameters during spatial–spectral token processing. The hyperparameter in GPE $\sigma = 7$ and We integrate the classification results of the three tokens by taking the mean.

The network was trained using an SGD optimizer with the default parameters set as: $learning_rate = 5e-3$, $momentum = 0.7$, $weight_decay = 1e-5$. We roughly set the epochs on the three datasets to 2000. All the implementations based on Pytorch backend [47] are performed on a desktop with one NVIDIA RTX3090 GPU.

C. Model Analysis

In order to comprehend the impact of different parameters on classification results and evaluate the effectiveness of the proposed modules in our CAT algorithm, we perform parameter sensitivity analysis, visual experiments, and ablation experiments in Sections III-C1–C6.

1) Parameter Sensitivity Analysis of Superpixel Region Number: In parameter sensitivity analysis experiments, we first compare the effect of different superpixel region sizes on the classification results. Huge superpixel size results in heightened internal heterogeneity, while tiny superpixel maintains internal consistency, but limiting the local receptive field. Consequently, the superpixel size should be within an appropriate range, which is related to the characteristics of the dataset itself. The experimental results are shown in Fig. 6. Overall, across the three datasets, when the number of superpixels increases, the classification OA of the CAT algorithm tends to rise initially at different sampling sizes, then levels off or slightly decreases. Moreover, On the IP dataset, as the number of superpixels increased, the classification OA initially improved up to a peak of 92.45% when the number of superpixels was set to 200 with 15×15 patches. However, further increasing the number of superpixels led to a decline in algorithm performance. Similar to the IP dataset, the CAT algorithm's OA initially improved as the number of superpixels increased on the PU dataset. When the number of superpixels reached 2000 with 15×15 patches, the OA reached 98.89%, indicating the effectiveness of the proposed CAT algorithm. However, as the number of superpixels continued to increase, the algorithm's performance tended to flatten out. In Houston 2013 dataset, when the number of superpixels is set to 24000, the OA for different patch sizes (5×5, 11×11, 15×15) are relatively close to each other, reaching 90.06, 90.75, and 89.73, respectively. This analysis helps determine the optimal superpixel region size that achieves the best classification accuracy and improves the spatial coherence of the classification map. The conclusion can be drawn from the above experiment that the optimal number of superpixels is 200 in IP, 2000 in PU and 24 000 in Houston 2013.

2) Parameter Sensitivity Analysis of Training Samples Percentages: Then we compare the test results under five different percentage of used training samples to evaluate the effectiveness of the MSRS module on PU dataset. The experimental results are shown in Fig. 7.

This experiment is conducted on the PU dataset, which involves four representative algorithms: 3D-CNN [11], RIAN [16], spectralformer [36], and SSFTT [39]. Additionally, the proposed CAT algorithm was compared with and without the MSRS module. In this experiment, "with_multi" refers to performing sampling three times, while "without_multi" refers to sampling only once. The results of the experiment revealed the following key observations. First, as the number of training samples decreases, the classification OA of all algorithms decreases. Second, the proposed CAT algorithm with the multiple sampling module demonstrated higher accuracy compared to the version without multiple sampling. This indicates that the MSRS effectively improves the classification accuracy of the CAT algorithm. Furthermore, when the number of training samples decreased slightly, the accuracy of the CAT algorithm with the multiple sampling module decreased at a slower rate compared to the version without multiple sampling. This suggests that the multiple sampling module helps to alleviate the problem of model overfitting, especially in cases with limited training samples.

3) Parameter Sensitivity Analysis of MSRS Times and Regions: Moreover, to analyze the gain of different repeated

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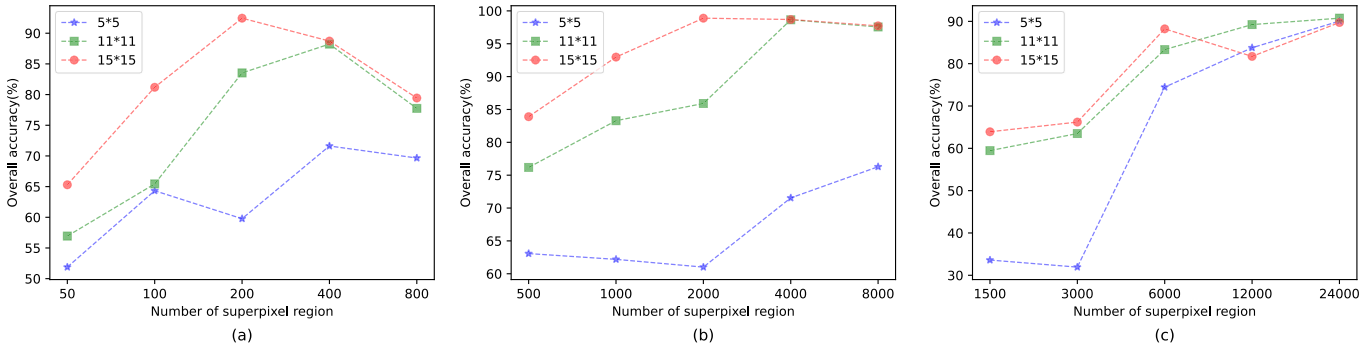

Fig. 6. Classification overall accuracies of CAT under different superpixel region sizes on three datasets. (a) IP. (b) PU. (c) Houston 2013. Three typical patch sizes were chosen for sampling data.

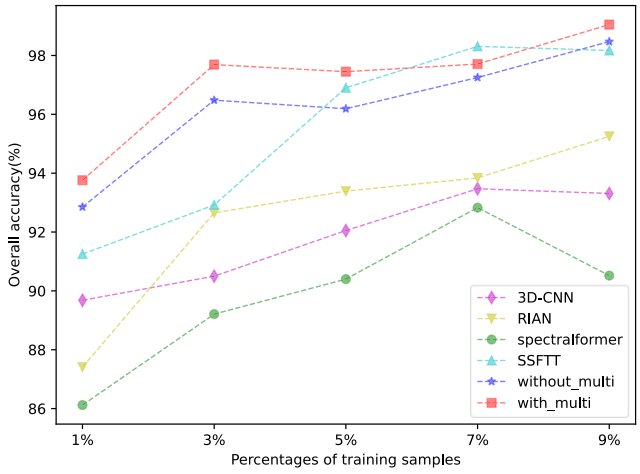

Fig. 7. Classification overall accuracies of six typical methods under different percentages of training samples on the PU dataset. (without_multi/with_multi means CAT without/with MSRS module).

TABLE II
ABLATION STUDY OF DIFFERENT SAMPLING TIMES

sampling times (t)	1	3	5	7	9
OA (%)	92.85	93.76	**94.08**	93.84	92.67
AA (%)	92.46	93.54	**93.59**	93.09	92.38
Kappa	0.9052	0.9162	**0.9181**	0.9161	0.9005
time cost (ms)	15.13	37.40	59.05	82.54	112.16

sampling times and regions on the classification effect, we implement quantitative experiments under five different sampling times and three different sampling regions while dealing with 1% training samples on PU dataset. The experimental results are shown in Tables II and III.

The classification accuracy (OA, AA, kappa) and sampling time of the models are evaluated in Table II. As the times of sampling increases, the classification accuracy shows a trend of increasing first and then decreasing. Specifically, the highest OA achieved by the models is 94.08%, the highest AA is 93.59%, and the highest kappa is 0.9181 when the number of samples is 5. On the other hand, the time cost grows rapidly with the increase in the number of sampling times. When sampling is performed nine times, the sampling time reaches 112 ms. This finding is consistent with the common observation that moderate sample expansion is helpful for model training but excessive introduction of homogeneous samples can lead to overfitting. Therefore, to balance the classification accuracy and the time cost, we set the number of sampling times to 3 in multiple stratified random module. It achieve comparable performance to the optimal five sampling times scenario while reducing the computational time required for data expansion. This data expansion strategy is useful when the available labeled samples are limited.

TABLE III
ABLATION STUDY OF DIFFERENT SAMPLING REGIONS

Homogeneous	First-order	Second-order	OA	AA	Kappa
✓	×	×	91.04	91.97	0.8770
✓	✓	×	**93.76**	**93.54**	**0.9162**
✓	✓	✓	91.07	90.20	0.8805

Table III shows the results with different sampling strategies in various regions. The default sampling times is 3 and the patch size is 15 × 15. Center region sampling means only sampling from the superpixel where the pixel to be classified belongs to. Adding with first-order neighborhood means 9×9 pixels sampled from the center region and the rest from first-order region. Similarly, introducing second-order neighborhood means 9×9 from the center region, $13\times13-9\times9$ from the first-order region, $15\times15-13\times13$ from the second-order region. The results show that the combination of the center region and first-order neighborhood can well balance the receptive field and the purity of the sample cube.

4) Parameter Sensitivity Analysis of GPE Variance: Ultimately, we compare the influence of GPE weights on the model's attention to the center, examining the impact of this parameter on the relevant features. The experimental results are shown in Table IV, and the data trend is shown in Fig. 8.

In this experiment, when the GPE parameter σ is smaller, it means that the distribution of attention weights becomes more concentrated around the central area. Therefore, when the patch size is small, the classification accuracy peak appears at σ equal to 1. As the patch size gradually increases, the range of the central area also needs to increase moderately. At this time, the peak value of classification accuracy gradually moves to the direction where sigma becomes larger. On the contrary, as the sigma approaches infinity, the GPE degenerates

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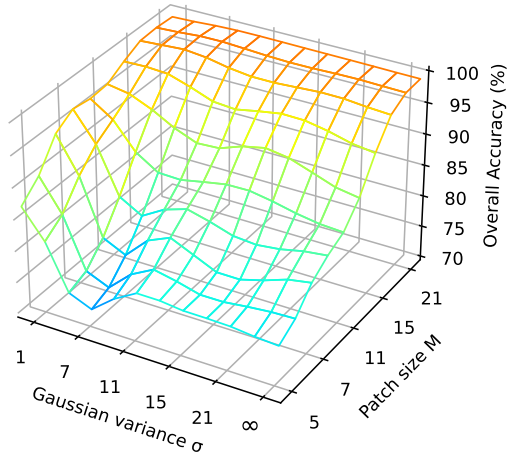

Fig. 8. Classification overall accuracies under different Gaussian variance σ and patch size M on the PU dataset.

TABLE IV
ABLATION STUDY OF DIFFERENT GAUSSIAN VARIANCE AND PATCH SIZE

M \ σ	1	7	11	15	21	∞
5× 5	**86.98**	74.56	77.93	78.51	78.31	77.61
7× 7	**94.82**	79.50	83.20	80.73	80.92	83.15
11 × 11	**95.27**	90.79	88.16	90.28	88.73	86.88
15× 15	98.37	**98.73**	97.14	97.23	97.45	96.64
21× 21	98.62	99.01	**99.10**	99.02	98.95	98.86

to sinusoidal positional embedding with periodic patterns that do not explicitly highlight any specific area, which leads to poor performance. This experiment showcases the capability of transformer-based model with GPE can focus on the central area of the HSI cube, enhancing the discriminative power of the model and improves classification accuracy. According to Table IV, we set $\sigma = ((M)^{1/2})/(2)$.

5) Visual Results of Center Attention: In the visual experiments, we project the preclassification features back into the original space and assigned pseudo color based on the associated weights. The visual results are shown in Fig. 9.

The first row represents the input image cubes, the second row displays the superpixel segmentation results along with the corresponding ground truth labels, and the third row showcases the superposition of the attention map onto the original images. We observe that regions sharing the same superpixel region as the central pixel exhibit higher weights, aligning with our expected hypothesis. This suggests that our CAT appropriately prioritizes features essential for classification.

6) Ablation Experiments of the Module Combination: In the ablation experiments, we insert each module into the basic framework one by one to assess the individual contribution of each module to the overall classification performance on the PU dataset. The experimental results are shown in Table V.

The transformer-based model with spatial–spectral token is treated as the base model for HSI classification on the PU dataset. The OA of the basic model is 96.28%. By utilizing the SSR module, the input data cube purity is enhanced and the classification results at the category boundary are improved. Therefore the OA is increased by 2.19% compared to the basic model. Then the introduction of MSRS module expands the training set size and increases the receptive field of the model, allowing it to capture more diverse and representative features. The OA is further increased by 0.39% compared to the model with SSR. Additionally, the addition of the GPE module enhances the model's attention on the classification pixel area, facilitating the extraction of separable features from the input data cubes. The OA is improved by an additional 0.24% compared to the former one. Ultimately, the complete model has the highest accuracy in all three indices, OA is 99.10%, AA is 99.03%, and Kappa is 0.9875.

TABLE V
ABLATION OF MODULE COMBINATION

SSR	MSRS	GPE	OA	AA	Kappa
×	×	×	96.28	97.18	0.9530
✓	×	×	98.47	98.45	0.9775
✓	✓	×	98.86	98.75	0.9843
✓	✓	✓	**99.10**	**99.03**	**0.9875**

By conducting these ablation experiments, we quantify the specific gain achieved by each module and understand their respective contributions to the overall classification effectiveness of the CAT model.

D. Experimental Comparison

To demonstrate the performance of the proposed CAT, the qualitative and quantitative classification results of CAT are compared with one traditional HSI classification method (SVM [5]), one CNN-based methods (3D-CNN [11]), one attention-based method (RIAN [16]), two GCN-based method (miniGCN [27], GTFN [48]), three transformer-based methods (ViT [43], spectralformer [36], and SSFTT [39]. These well-established models are representative and highly related to the proposed method CAT. Some strongly related articles are published in recent one year. This comparison helps in understanding the unique contributions of CAT and its potential advantages over existing methods. The corresponding analysis and discussion of the comparison results are reported in this section.

The first experiment is reported on the Indian Pines dataset. As shown in Table I, we randomly select 50 samples from each class for training and the remaining samples for testing. For categories with a total number of samples less than 50, such as Alfalfa, Grass-pasture-mowed, and Oats, 15 samples are selected for training. The accuracy in each class, OA, AA, and Kappa of the proposed CAT as well as the compared methods are reported in Table VI. The highest value in each row is highlighted in bold. We observe that the conventional method SVM and transformer-based method ViT (without the hyperspectral design module) have similar OA and kappa on IP dataset. This indicates that the pertinence and rationality of the structural design are important when applying transformer-based methods to hyperspectral classification tasks. Otherwise, the transformer-based method

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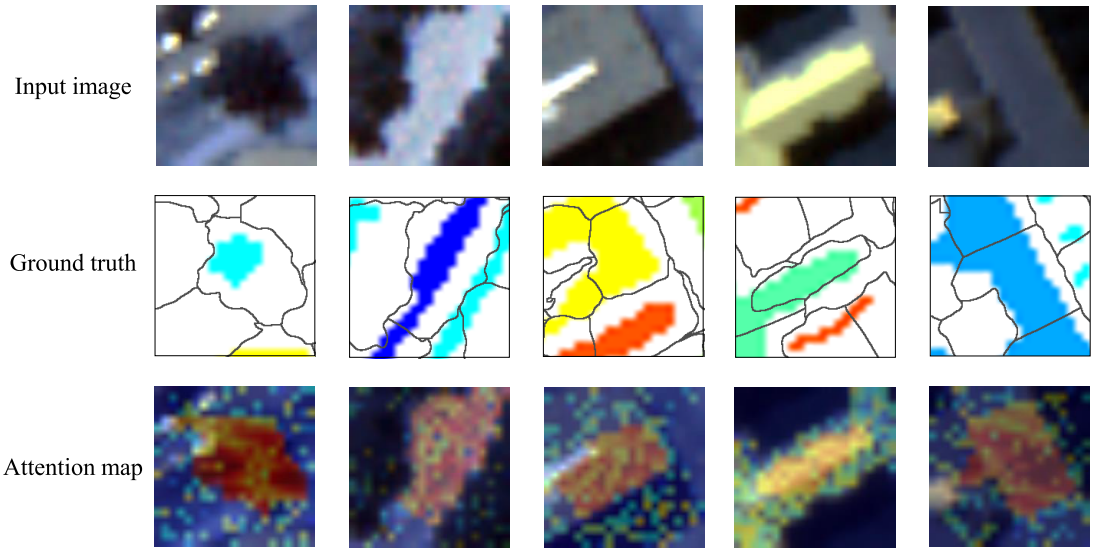

Fig. 9. Visual results for the center attention. Five typical ground categories and their respective attention maps have been displayed.

TABLE VI
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE INDIAN PINES DATASET (OPTIMAL RESULTS ARE BOLDED)

class	SVM	3D-CNN	miniGCN	GTFN	RIAN	ViT	spectralformer	SSFTT	ours
1	82.05	66.26	48.72	94.45	76.92	79.49	79.49	**94.87**	**94.87**
2	67.34	71.94	72.54	**94.87**	88.58	53.25	70.52	84.47	84.75
3	67.86	**97.28**	55.99	80.27	79.97	66.20	81.89	89.92	92.47
4	93.48	95.08	92.93	89.80	77.72	86.41	91.30	92.39	**99.46**
5	94.63	88.09	92.62	**100.00**	86.13	89.71	95.53	76.51	93.29
6	88.52	**98.18**	94.98	76.51	82.21	87.66	85.51	97.99	97.85
7	90.91	75.38	72.73	98.79	**100.00**	90.91	**100.00**	90.91	90.91
8	94.76	56.29	98.63	97.42	**99.54**	89.98	99.32	99.09	99.09
9	**100.00**	78.01	80.00	97.78	**100.00**	80.00	**100.00**	**100.00**	**100.00**
10	73.86	**100.00**	64.71	90.91	89.11	72.22	81.81	92.48	90.52
11	52.07	90.59	68.78	**99.09**	88.17	66.00	75.48	92.80	95.24
12	72.70	90.30	69.33	**100.00**	80.85	57.09	73.76	91.84	90.25
13	98.77	**100.00**	98.77	90.20	92.59	97.53	98.77	**100.00**	92.59
14	86.17	74.36	87.78	91.23	98.87	87.62	93.17	93.57	**98.95**
15	71.82	**100.00**	50.00	88.83	95.45	63.94	78.48	98.79	98.79
16	95.56	**100.00**	**100.00**	**100.00**	97.78	95.56	**100.00**	97.78	97.78
OA	72.36	75.48	75.11	90.47	88.74	71.86	81.76	91.65	**93.69**
AA	83.16	86.36	78.03	93.13	89.62	78.97	87.81	93.34	**94.80**
Kappa	0.6888	0.7240	0.7164	0.8907	0.8711	0.6804	0.7919	0.9042	**0.9275**

will be less effective. CNNs (3D-CNN) naturally capture neighborhood semantic information through their convolutional design, leading to a higher classification accuracy compared to conventional methods. The ability to consider local context and extract spatial features greatly contributes to their superior performance. The GCNs (miniGCN, GTFN) guide the selection of sample points by leveraging spatial relationships. These approaches help capture spectral–spatial information effectively, resulting in comparable performance to CNN-based methods. The Attention mechanism RIAN enables the model to allocate more attention to informative areas, leading to enhanced classification accuracy. Furthermore, Spectralformer and SSFTT incorporate design elements tailored to HSI classification, such as patch-wise spectral token in Spectralformer and the dual branch architecture involving the space and spectrum tokens in SSFTT. It is apparent that even though there may be certain categories where other methods achieve the best classification accuracy individually, the advantage of our method CAT lies in its overall balanced performance across all categories. Therefore, when only five categories achieve the best classification accuracy, our OA and kappa coefficients are still 2.04 and 0.0233 higher than the second place SSFTT, which is also based on the transformer architecture.

The second experiment is evaluated on the Pavia University dataset. As shown in Table I, the same number of samples as mentioned in [36] is used to split the train and test sets. The accuracy in each class, OA, AA, and Kappa of the proposed CAT as well as the compared methods are reported in Table VII. The highest value in each row is highlighted in bold. The Pavia University dataset has fewer categories (nine categories) compared to the Indian Pines dataset and the category differences are relatively large, which makes it easily separable for classification. It is interesting to observe that most algorithms achieve an overall classification accuracy (AA) above 85%. The high accuracy (over 90%) observed

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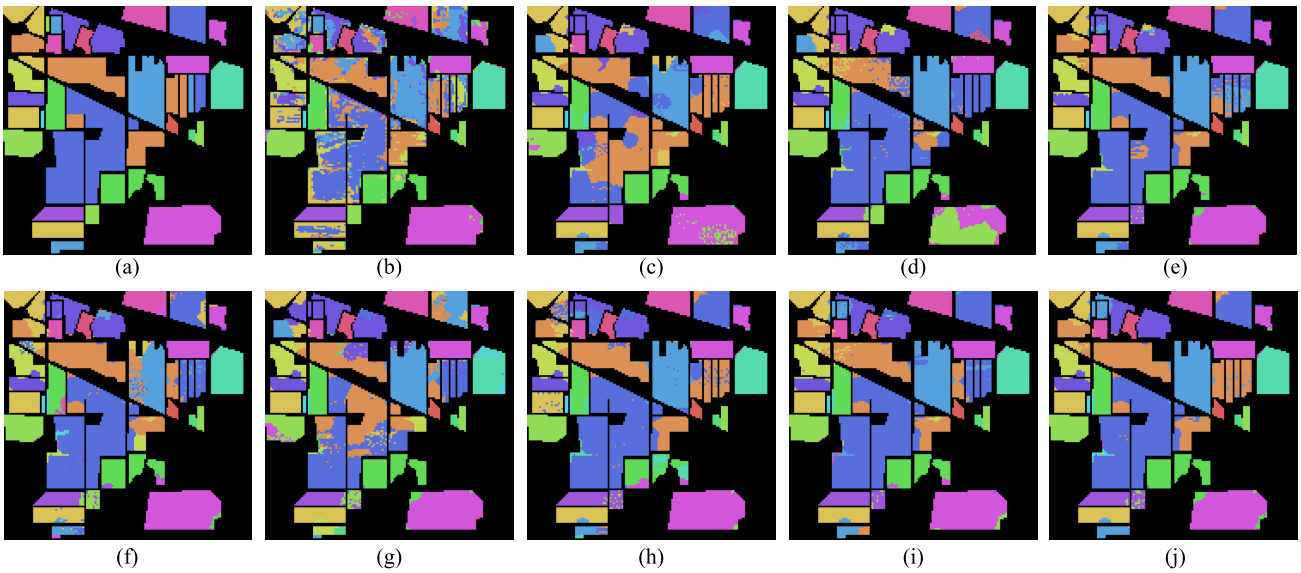

Fig. 10. Classification maps obtained by the different tested methods for the Indian Pines dataset. (a) GT. (b) SVM. (c) 3D-CNN. (d) miniGCN. (e) GTFN. (f) RIAN. (g) ViT. (h) Spectralformer. (i) SSFTT. (j) Ours.

for the Trees, Painted metal sheets and Shadows categories can be attributed to the distinct spectral characteristics of these categories. In contrast to the Indian Pines dataset, where multiple categories are crops and may exhibit similar spectral features, the Pavia University dataset with its distinct categories allows for higher accuracy in classifying these particular categories. While GCN and Vit offer flexibility in sample selection, their accuracy is similar or lower compared to the CNN-based methods (3DCNN), with an approximately 5% point difference in OA, AA, and kappa indicators for Vit. This suggests that the specific design choices and mechanisms used in GCN and Vit may not be as effective in capturing the spectral and spatial characteristics of the hyperspectral data in this dataset. RIAN (Attention-based method) leverages the advantages of the attention mechanism, improving attention on the same area as the central pixel through attention. This approach indirectly realizes sample selection and shows a strong classification potential. The structural design of RIAN is also better aligned with the requirements of hyperspectral classification, resulting in improved performance compared to CNN-based methods. Other Transformer-based algorithms also outperformed the CNN-based method. Spectralformer achieved a kappa indicator of 0.87, SSFTT achieved 0.97, and CAT achieved 0.99. This demonstrates the strong classification potential of self-attention mechanisms in combination with hyperspectral data. The improved performance of these methods suggests that considering sample selection and the rationality of the network structure design simultaneously can enhance the overall classification effect.

The third experiment is conducted on the Houston 2013 dataset. As shown in Table I, we randomly select nearly 200 samples from each class for training and the remaining samples for testing, which is the same number of samples as mentioned in [36] is used to split the train and test sets The accuracy in each class, OA, AA, and Kappa of the proposed CAT as well as the compared methods are reported in Table VIII. The highest value in each row is highlighted in bold. In Houston 2013 dataset, the scattered nature of labeled samples and the limited spatial continuity of the ground features indeed present unique challenges for classification compared to datasets like Pavia University and Indian Pines. Compared with CNN-based methods, the flexibility in sample selection allows our method to deal with categories that exhibit nonblock characteristics in spatial distribution. By leveraging this flexibility, our CAT demonstrates its advantage to handle challenging scenarios where traditional CNN-based methods may struggle due to their reliance on block spatial features. Compared with attention-based method RIAN, our CAT and the RIAN have their best performance in different categories. This suggests that the two approaches have different strengths in capturing and utilizing the relevant spatial and spectral information for classification. Our observation that CAT performs better overall than RIAN (approximately 6% higher on OA) indicates that direct sample selection is more effective than the attention weight distribution approach. Compared with the transformer-based methods, we observe that the effectiveness of the structure design plays a significant role in classification results. The position encoding scheme in our method highlights the importance of the central area for classification. The utilization of multiple sampling in our method helps mitigate the risk of overfitting. Hence our CAT results in superior performance compared to spectralformer and SSFTT on OA, AA, and kappa index.

For qualitative evaluation, we provide meaningful insights into the performance of different algorithms on the three datasets, Indian pines, Pavia university, and Houston 2013. As shown in Figs. 10–12, we observe that SVM, which does not fully utilize spatial information, may result in noisy visualization and less smooth in grounds distribution. This is because SVM primarily focuses on spectral features and may not effectively capture the spatial relationships between neighboring pixels. On the other hand, CNN-based

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TABLE VII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE PAVIA UNIVERSITY DATASET (OPTIMAL RESULTS ARE BOLDED)

class	SVM	3D-CNN	miniGCN	GTFN	RIAN	ViT	spectralformer	SSFTT	ours
1	74.32	94.24	93.36	93.64	90.44	91.18	84.21	97.13	**97.75**
2	64.13	90.67	91.19	94.45	96.63	77.28	90.66	98.66	**99.54**
3	20.55	95.13	95.07	98.03	93.50	92.63	98.90	99.19	**99.83**
4	93.75	94.76	92.02	96.68	90.94	90.94	93.14	93.44	**97.99**
5	99.01	98.90	98.90	98.99	98.71	98.44	98.81	99.08	**99.82**
6	75.79	99.73	95.23	99.29	98.82	90.50	90.30	99.96	**100.00**
7	85.52	94.80	95.84	98.65	98.13	98.34	98.13	**99.58**	**99.58**
8	92.33	92.51	86.02	89.38	93.95	88.91	89.82	97.24	**97.37**
9	99.12	95.24	97.15	95.38	95.24	91.98	93.07	99.05	**99.18**
OA	71.81	93.31	92.24	95.02	95.25	84.90	90.52	98.17	**99.05**
AA	87.28	95.11	93.86	96.05	95.15	91.13	93.00	98.15	**99.01**
Kappa	0.6427	0.9105	0.8959	0.9328	0.9355	0.8028	0.8725	0.9751	**0.9870**

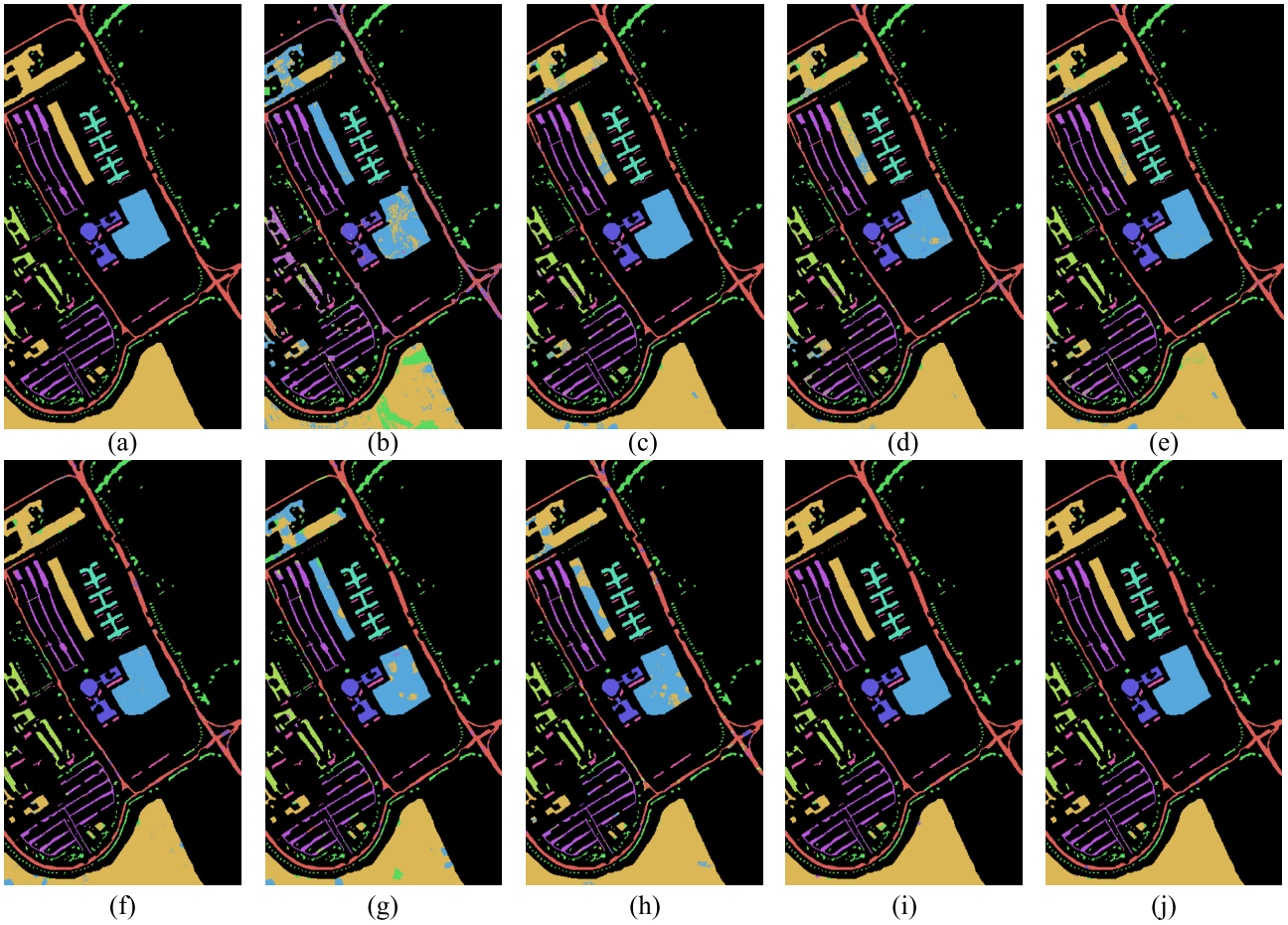

Fig. 11. Classification maps obtained by the different tested methods for the Pavia University dataset. (a) GT. (b) SVM. (c) 3D-CNN. (d) miniGCN. (e) GTFN. (f) RIAN. (g) ViT. (h) Spectralformer. (i) SSFTT. (j) Ours.

methods, such as 3D-CNN, leverage their ability to extract local receptive field features, resulting in smoother grounds distribution. The convolutional layers of CNN models allow them to capture spatial patterns and contextual information. However, the similarity of certain category features, such as the high spectral similarity of crops in the Indian Pines dataset, can still lead to misclassification cases, which can be observed in the results of 3D-CNN. GCN and RIAN both employ different strategies to enhance the discriminative features extraction capabilities for similar samples, leading to improved classification results. GCN utilizes graph-based methods to construct a map of the input samples, taking into account the relationships between neighboring samples. While RIAN leverages attention mechanisms to weigh the importance of features between samples. Since the attention mechanism effectively captures separable characteristics of samples, RIAN outperforms other CNN-based methods in terms of distinguishing similar categories and demonstrates excellent performance in visualization. Furthermore, we observe that the visualization results of transformer-based methods, such

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TABLE VIII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON 2013 DATASET (OPTIMAL RESULTS ARE BOLDED)

class	SVM	3D-CNN	miniGCN	GTFN	RIAN	ViT	spectralformer	SSFTT	ours
1	83.00	84.71	**98.39**	56.12	83.00	82.81	81.86	79.96	88.13
2	98.40	99.34	92.11	96.33	83.74	96.62	**100.00**	97.27	90.04
3	99.60	84.55	99.60	94.26	89.70	99.80	95.25	99.60	**100.00**
4	98.48	98.01	96.78	79.17	92.52	**99.24**	96.12	84.94	89.49
5	97.82	97.82	97.73	97.44	**99.81**	97.73	99.53	98.67	98.20
6	90.91	93.01	95.10	90.91	**100.00**	95.10	94.41	97.90	99.30
7	90.39	86.29	57.28	92.63	87.59	76.77	83.12	94.31	**98.97**
8	40.46	76.26	68.09	78.73	73.22	55.65	76.73	95.06	**98.01**
9	41.93	84.23	53.92	87.25	81.21	67.42	79.32	94.33	**95.94**
10	62.64	74.32	77.41	86.97	68.15	68.05	78.86	96.81	**98.65**
11	75.43	82.35	84.91	**96.02**	89.85	82.35	88.71	82.64	87.38
12	60.04	77.71	77.23	64.27	89.15	58.50	87.32	88.09	**94.33**
13	49.47	81.05	50.88	90.53	92.28	60.00	72.63	93.33	**98.95**
14	98.79	88.66	98.38	99.60	**100.00**	98.79	**100.00**	98.79	98.79
15	97.46	84.57	98.52	96.83	**100.00**	98.73	99.79	97.46	62.16
OA	76.91	86.04	81.71	85.08	86.30	80.41	88.01	92.09	**93.21**
AA	78.99	86.19	83.09	87.14	88.68	82.50	88.91	**93.28**	93.22
Kappa	0.7494	0.8598	0.8062	0.8381	0.8514	0.7949	0.8732	0.9141	**0.9263**

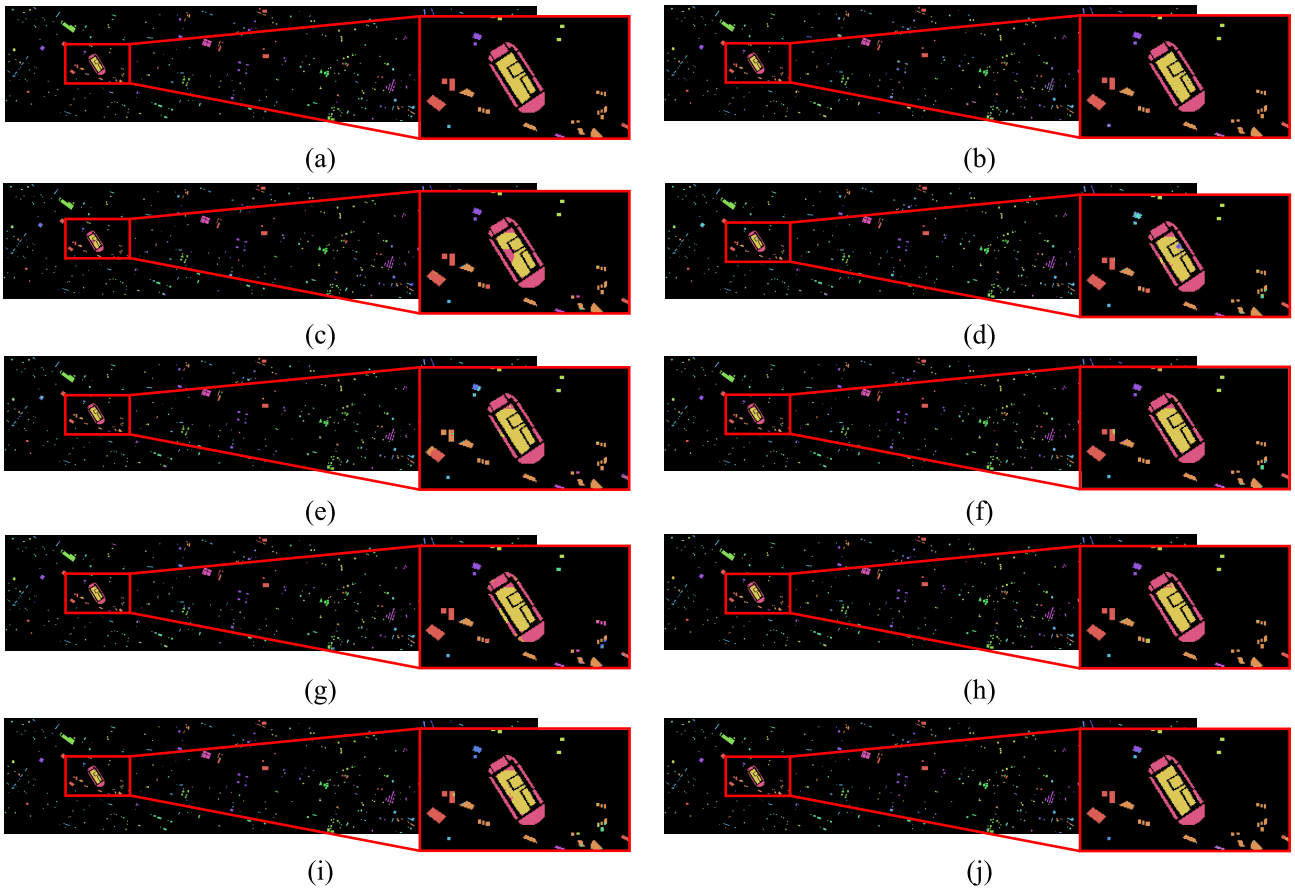

Fig. 12. Classification maps obtained by the different tested methods for the Houston 2013 dataset. (a) GT. (b) SVM. (c) 3D-CNN. (d) miniGCN. (e) GTFN. (f) RIAN. (g) ViT. (h) Spectralformer. (i) SSFTT. (j) Ours.

as ViT, spectralformer, and SSFTT, are quite different from the CNN-based methods. ViT directly uses spectral vectors as tokens for feature mining, which may not fully capture the specific information contained within the individual bands of the ground object. Spectralformer utilizes spectral band segments to calculate cross-attention, enabling it to better mine spectral information. By considering the inter-band relationships, spectralformer improves the visualization effect compared to ViT. SSFTT, with its simultaneous consideration of spatial and spectral information, further enhances the feature mining capability for HSIs. By leveraging both spatial and spectral information, SSFTT can extract more discriminative features and reduce noise within categories, resulting in smoother classification results compared to spectralformer. Our method

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CAT combines the advantages of spatial–spectral information mining and effective sample extraction through superpixel sampling. By selecting samples from homogeneous regions, CAT ensures that the training set represents the various classes in a more balanced and representative manner. CAT is able to achieve superior performance in the visualization results.

To sum up, based on the analysis of the quantitative and qualitative results, our CAT method demonstrates superior performance compared to the state-of-the-art methods in HSI classification. This superiority is evident in two key aspects: 1) Best classification accuracy on overall test HSI datasets. 2) Smooth effect in visual classification maps. By excelling in both the quantitative evaluation metrics and the qualitative visualization results, CAT demonstrates its superiority in HSI classification tasks. The comprehensive analysis supports the conclusion that CAT can reliably and accurately classify land covers, irrespective of their category, class number, or spatial distribution. This highlights the importance of a comprehensive and targeted approach in HSI classification.

IV. Conclusion and Future Works

In conclusion, this article propose a CAT framework for HSI classification. The key contributions of this work are summarized as follows. The SSR mechanism removes heterogeneous samples at category boundaries and enhances the purity of the image cube. The MSRS mechanism generates various training samples without additional labels and alleviate the overfitting problem. The Spatial–Spectral Token Generation mechanism and GPE method both enhance spatial–spectral separable features mining ability of CAT. These modules are shown to improve HSI classification results in the above experiments.

The findings of this work highlight the importance of incorporating spatial information, handling heterogeneous samples, mitigating overfitting, and enhancing spatial–spectral separable features mining in the context of HSI classification. Future research can build upon these findings to further improve the performance and applicability of the CAT framework in real-world HSI classification scenarios.

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